MARKING SCHEME

Model Paper — 02

Mathematics I

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Notes (for teachers):

- Part A: 2 marks per question (1 mark for the correct method / working, 1 mark for the correct final answer). Total 50.
- Part B: 10 marks per question, total 50. The marks for each step are given in the right-hand column and add up to the marks printed against each sub-part.
- Give full marks for correct alternative methods. Accept reasonable readings from graphs and drawings (ranges are shown).

Part A — Answers

1. $8 \times 18 = 144$ worker-days; workers needed for 12 days $= 144 \div 12 = 12$; extra $= 12 - 8$

Final answer: 4 workers

(2 marks)

2. Through E draw a line parallel to AB: $A\hat{E}C = B\hat{A}E + E\hat{C}D$ (alternate angles); $125° = B\hat{A}E + 70°$

Final answer: 55°

(2 marks)

3. LCM $= 12x$: $\frac{9 - 10 + 8}{12x}$

Final answer: $\frac{7}{12x}$

(2 marks)

4. Faces of a right prism are rectangles (and the two triangular ends); a square or an unequal-sided rectangle can be a lateral face, but a parallelogram with no right angle cannot

Final answer: (iii)

(2 marks)

5. Magazines sector $= 360° - (180° + 90° + 20°) = 70°$; $90° \leftrightarrow 54$ books, so $70° \leftrightarrow 54 \div 90 \times 70$

Final answer: 42 magazines

(2 marks)

6. DC $\perp$ AB (line from the centre to the mid-point of a chord); $A\hat{D}B + A\hat{E}B = 180°$ (opposite angles of a cyclic quadrilateral) $\Rightarrow A\hat{D}B = 80°$; DC bisects $A\hat{D}B$ ($\triangle ADC \equiv \triangle BDC$) $\Rightarrow C\hat{D}B = 80° \div 2$

Final answer: 40°

(2 marks)

7. $4(x^2 - 9)$

Final answer: $4(x + 3)(x - 3)$

(2 marks)

8. $358 = 3.58 \times 10^2$; $\lg 358 = 2 + \lg 3.58 = 2 + 0.5539$

Final answer: 2.5539

(2 marks)

9. a marked at C between CA and CB; b marked at B between the horizontal through B and BC; the horizontals are parallel, so a and b are alternate angles

Final answer: $a = b$

(2 marks)

10. Curved surface $= 2\pi r(5r) = 10\pi r^2$; one base $= \pi r^2$

Final answer: 10 : 1

(2 marks)

11. 16 equally likely outcomes; sum 6: (1,5), (5,1), (2,4), (4,2) = 4 outcomes; $\frac{4}{16}$

Final answer: $\frac{1}{4}$

(2 marks)

12. $AD = DB \Rightarrow D\hat{A}B = D\hat{B}A = (180° - 40°) \div 2 = 70°$; $D\hat{B}C = 180° - 70° = 110°$; $BD = BC \Rightarrow x = (180° - 110°) \div 2$

Final answer: $x = 35°$

(2 marks)

13. $A = \{3, 6, 9, 12\}$, $B = \{2, 4, 6, 8, 10, 12\}$; $A \cap B = \{6, 12\}$; $A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12\}$ so $n(A \cup B) = 8$; $n(A) = 4$

Final answer: (ii)

(2 marks)

14. $6x^2y^3 \times 4x^3y = 24x^5y^4$; $\frac{24x^5y^4}{12xy^2}$

Final answer: $2x^4y^2$

(2 marks)

15. $r = (-6) \div (-2) = 3$; $T_5 = T_3 \times r^2 = -6 \times 9$

Final answer: −54

(2 marks)

16. $B\hat{D}C = B\hat{A}C = 38°$ (angles in the same segment); $A\hat{D}C = 2 \times 38° = 76°$; $C\hat{B}E = A\hat{D}C$ (exterior angle of a cyclic quadrilateral)

Final answer: 76°

(2 marks)

17. HCF of 8, 12, 20 = 4; lowest powers a^2, b

Final answer: $4a^2b$

(2 marks)

18. Volume of prism = $154h$; $\pi r^2 h = 154h \Rightarrow \frac{22}{7} r^2 = 154 \Rightarrow r^2 = 49$

Final answer: $r = 7$ cm

(2 marks)

19. AE = $\sqrt{(8^2 + 6^2)} = 10$, so EC = 10; BC = 2 × DE = 12 (mid-point theorem); DB = 8; perimeter = 12 + 10 + 6 + 8

Final answer: 36 cm

(2 marks)

20. Gradient = $\frac{7 - 1}{5 - 2} = 2$; $y - 1 = 2(x - 2)$

Final answer: $y = 2x - 3$

(2 marks)

21. $D\hat{A}B = B\hat{C}D = 70°$ (opposite angles of a parallelogram); $A\hat{E}B = 90°$ (angle in a semicircle); $A\hat{B}E = 180° - 90° - 70° = 20°$; $E\hat{B}F = 180° - 20°$

Final answer: 160°

(2 marks)

22. Amount at the start of year 2 = $2\,160 \div \frac{8}{100} = 27\,000$; amount invested = $27\,000 \div 1.08$

Final answer: Rs. 25 000

(2 marks)

23. AC ∥ BD ⇒ $A\hat{C}B = C\hat{B}D = 50°$ (alternate angles); angle at the centre = 2 × angle at the circumference ⇒ $A\hat{E}B = 50° \div 2$

Final answer: 25°

(2 marks)

24. $2x > 1 - 9 \Rightarrow 2x > -8 \Rightarrow x > -4$

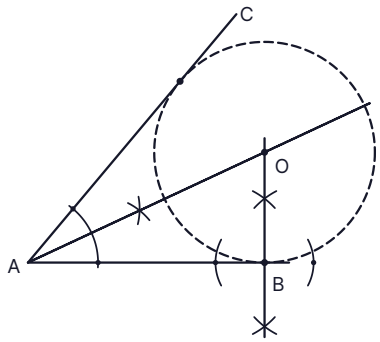
Final answer: −3, −2, −1

(2 marks)

25. Draw the bisector of angle A through the arc intersection (1 mark); draw the perpendicular to AB at B through the two arc intersections (1 mark); they meet at O

Final answer: O = intersection of the angle bisector at A and the perpendicular at B (OB = distance to AC)

(2 marks)



Completed construction: bisector of angle A, perpendicular to AB at B, meeting at O (the dashed circle is optional)

Part B — Answers and mark allocation

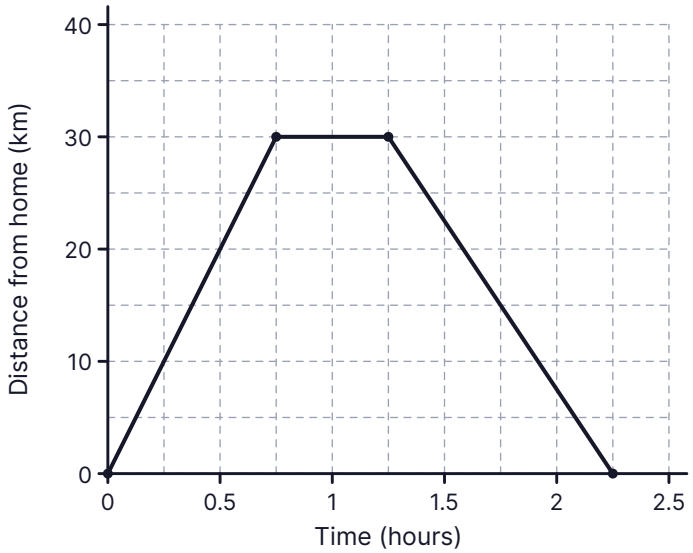
Question 1

(i)	
$1 - \frac{1}{4} = \frac{3}{4}$	**1**
(ii)	
Spent on other expenses = $\frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$	**1**
$\frac{3}{4} - \frac{3}{10} = \frac{9}{20}$ of the money received	**1**
(iii)	
Money received = 55 125 ÷ $\frac{9}{20}$ = Rs. 122 500	**1**
Food = 122 500 ÷ 4 = Rs. 30 625 ; other expenses = $\frac{3}{10}$ × 122 500 = Rs. 36 750	**1**
(iv)	
Let the salary be Rs. S: S − 10% (S − 100 000) = 122 500	**1**
0.9 S + 10 000 = 122 500 ⇒ 0.9 S = 112 500	**1**
S = Rs. 125 000	**1**
(v)	
$\frac{30\ 625}{125\ 000} \times 100$	**1**
= 24.5%	**1**

Total: 10

Question 2

(a)(i)	
Horizontal line from (0.75, 30) to (1.25, 30) (stay of 30 minutes)	**1**
Straight line from (1.25, 30) to (2.25, 0) (return in 1 hour), joined to the first part	**1**
(a)(ii)	
Speed = 30 km ÷ 1 h = 30 km h^{-1}	**1**
(a)(iii)	
15 km from home at time 1.75 h on the graph; so 1 h 45 min after leaving (accept 1.7 – 1.8 h)	**1**
(b)(i)	
Arc AEB = $\frac{1}{2} \times 2 \times \frac{22}{7} \times 7 = 22$ cm	**1**
Perimeter = 22 + 55 + 14 + 55	**1**
= 146 cm	**1**
(b)(ii)	
Area of rectangle = 14 × 55 = 770 cm^2	**1**
Area of semicircle = $\frac{1}{2} \times \frac{22}{7} \times 7 \times 7 = 77$ cm^2	**1**
770 ÷ 77 = 10 times	**1**



Completed graph: (0, 0) – (0.75, 30) – (1.25, 30) – (2.25, 0)

Total: 10

Question 3

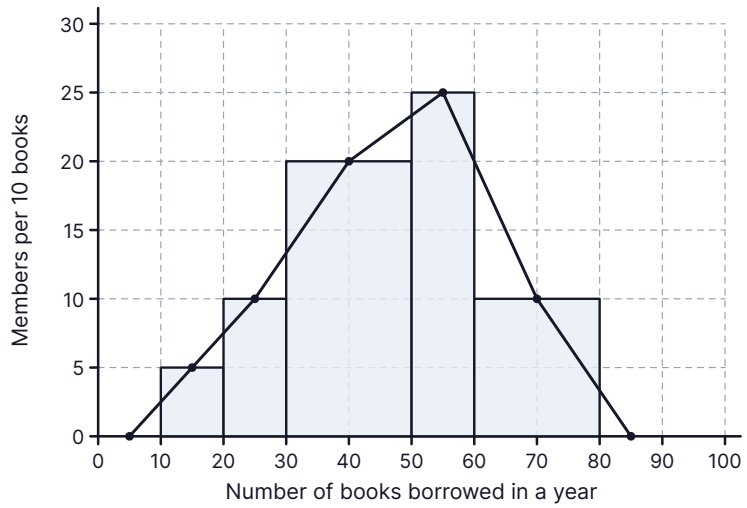
(i)	
Rates = $\frac{15}{100} \times 80\,000$	**1**
= Rs. 12 000	**1**
(ii)	
Annual rent = 10 000 × 12 = Rs. 120 000 ; maintenance = $\frac{10}{100} \times 120\,000$ = Rs. 12 000	**1**
120 000 − 12 000 − 12 000 = Rs. 96 000	**1**
(iii)(a)	
Number of shares = 6 000 ÷ 4	**1**
= 1 500	**1**
Amount invested = 1 500 × 80 = Rs. 120 000	**1**
(iii)(b)	
Extra money added = 120 000 − 96 000 = Rs. 24 000	**1**
$\frac{24\,000}{120\,000} \times 100$	**1**
= 20%	**1**

Total: 10

Question 4

(i)	
Frequency = height × (class width ÷ 10)	**1**
Readings 5, 10, 20, 25, 10; frequencies: 20 – 30 → 10 and 50 – 60 → 25	**1**
30 – 50 → 20 × 2 = 40 and 60 – 80 → 10 × 2 = 20	**1**
(ii)	
Bars 10 – 20 (height 5) and 20 – 30 (height 10) correct	**1**
Bar 30 – 50 (width 20, height 20) correct	**1**
Bars 50 – 60 (height 25) and 60 – 80 (height 10) correct; tops of the bars match the polygon points	**1**
(iii)	
Cumulative frequencies 5, 15, 55, 80, 100 (all correct 2 marks; three or four correct 1 mark)	**2**
(iv)	
Members with 50 or more books = 25 + 20 = 45 (or 100 − 55)	**1**
Probability = $\frac{45}{100} = \frac{9}{20}$	**1**

Number of books	**10 – 20**	**20 – 30**	**30 – 50**	**50 – 60**	**60 – 80**
Number of members	5	10	40	25	20
Cumulative frequency	5	15	55	80	100

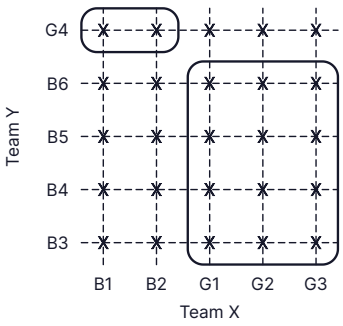


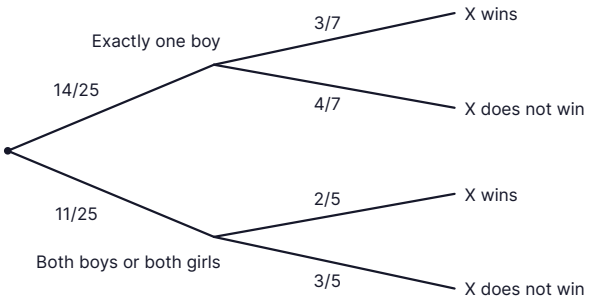
Completed histogram

Total: 10

Question 5

(i)	
All 25 outcomes (5 × 5) marked with a cross	**2**
(ii)	
Closed curve around the 2 outcomes (B1, B2) with G4	**1**
Closed curve around the 12 outcomes (G1, G2, G3) with (B3, B4, B5, B6); in all 14 outcomes	**1**
P(exactly one boy) = $\frac{14}{25}$	**1**
(iii)	
First branches: $\frac{14}{25}$ (exactly one boy) and $\frac{11}{25}$ (both boys or both girls)	**1**
After "exactly one boy": X wins $\frac{3}{7}$, does not win $\frac{4}{7}$	**1**
After "both boys or both girls": X wins $\frac{2}{5}$, does not win $\frac{3}{5}$	**1**
(iv)	
$\frac{14}{25} \times \frac{4}{7} + \frac{11}{25} \times \frac{3}{5} = \frac{8}{25} + \frac{33}{125}$	**1**
$= \frac{73}{125}$ (check: P(X wins) = $\frac{52}{125}$)	**1**





Total: 10